Técnicas Avanzadas de Data Mining y Sistemas Inteligentes

Maestría en Informática
Escuela de Posgrado
Pontificia Universidad Católica del Perú

2024-2

César A. Beltrán Castañón



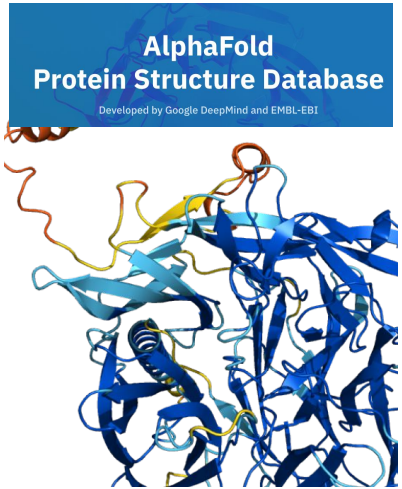
- **Peru Section:** (2023 – 2024) Section Chair
  (2021 – 2022) Awards Chair
- **Computer Society:** (2019-2020) Chapter Chair
  (2015-2018) Chapter Vice-chair

Full Professor and Senior Researcher at the Pontificia Universidad Católica del Perú, he is founder and leader of the Scientific Group on Artificial Intelligence (IA-PUCP). Doctor in Bioinformatics and Master in Computer Science, both from the University of Sao Paulo (USP), Brazil; He completed a Post-doctoral stay at Texas A&M University (USA); He has a degree in Systems Engineering from the Catholic University of Santa María (Arequipa).

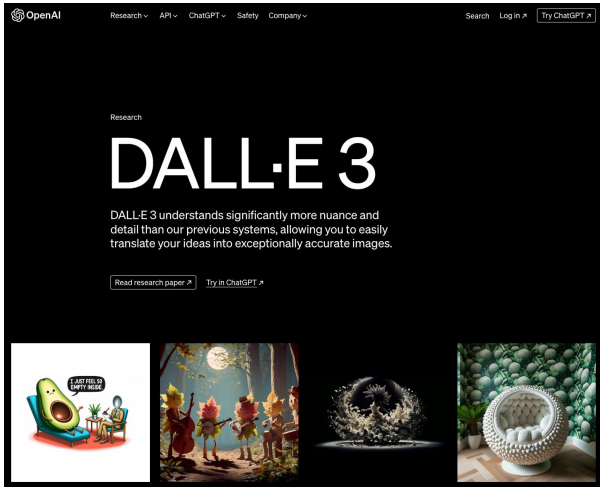
IEEE Senior Member
cbeltran@ieee.org

Aplicaciones de IA

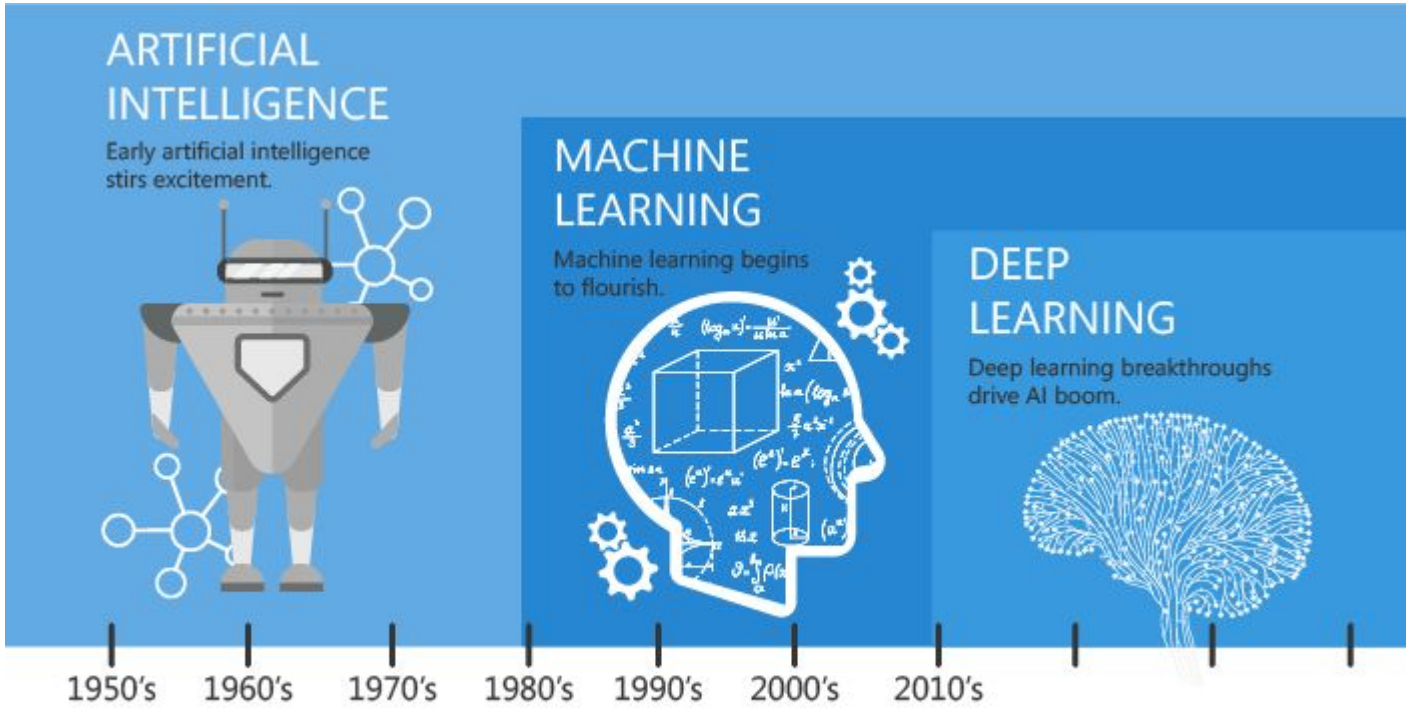




https://alphafold.ebi.ac.uk/



https://openai.com/dall-e-3



Since an early flush of optimism in the 1950's, smaller subsets of artificial intelligence - first machine learning, then deep learning, a subset of machine learning - have created ever larger disruptions.

https://www.linkedin.com/pulse/artificial-intelligence-ai-vs-machine-learning-deep-natarajan-siva/

Machine Learning

Traditional Programming



Traditional Programming

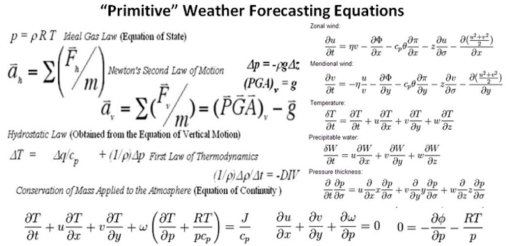


Machine Learning



Traditional Programming



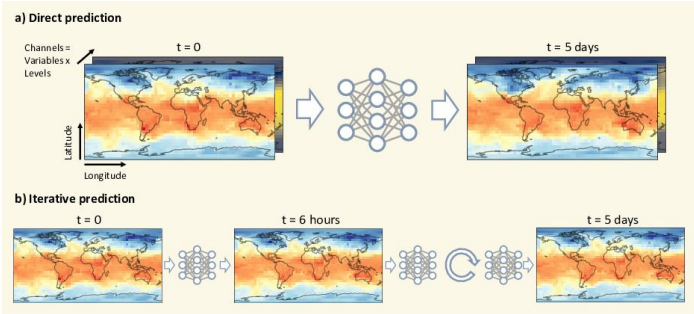


https://www.iweathernet.com/educational/history-weather-forecasting

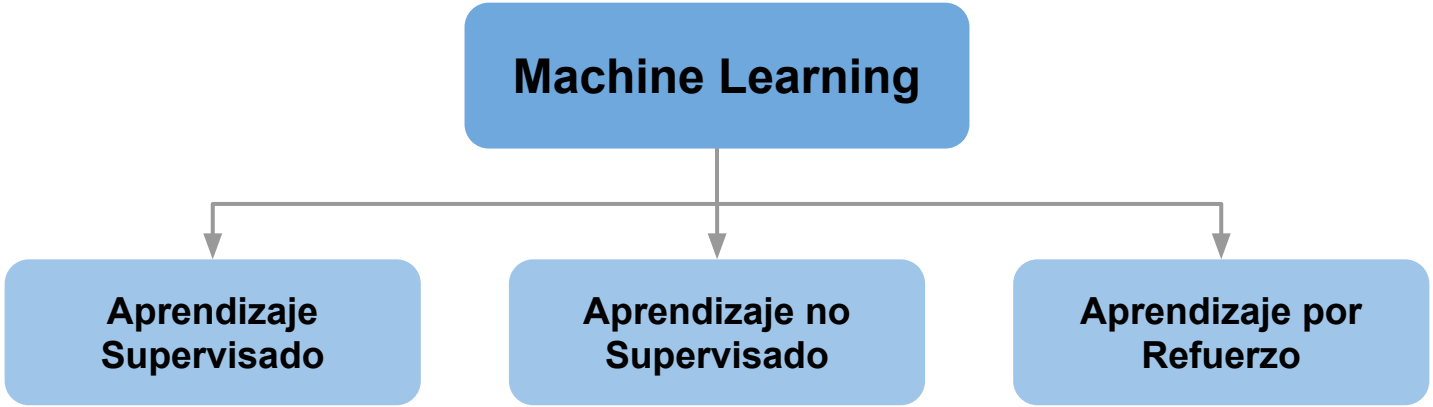
Machine Learning



Los algoritmos de machine learning usan métodos computacionales para aprender directamente de la data, sin estar explícitamente programados.



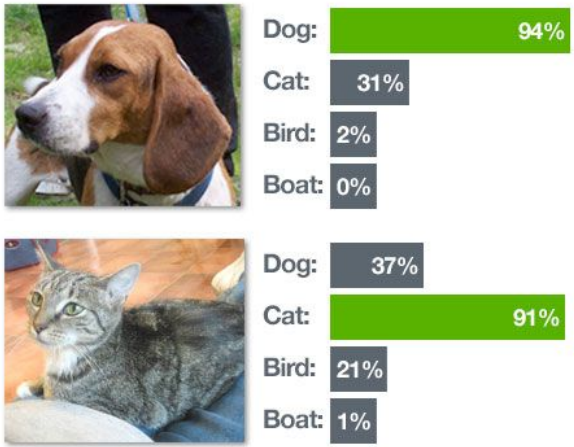
https://raspstephan.github.io/blog/weatherbench/#

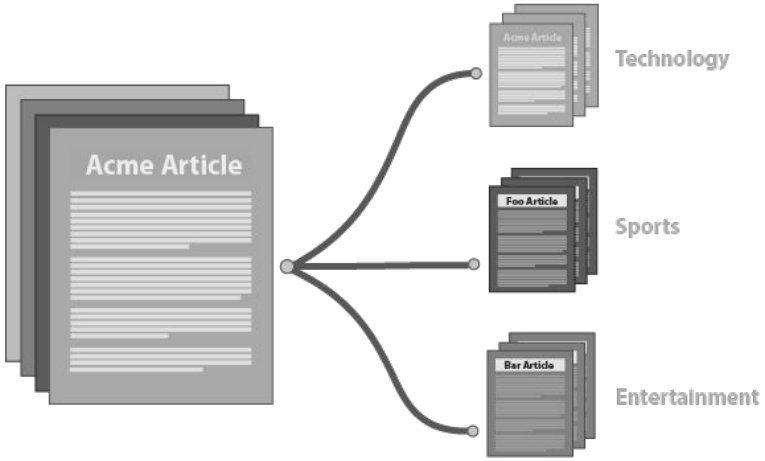

Machine Learning
Aprendizaje Supervisado
Aprendizaje no Supervisado
Aprendizaje por Refuerzo

Machine

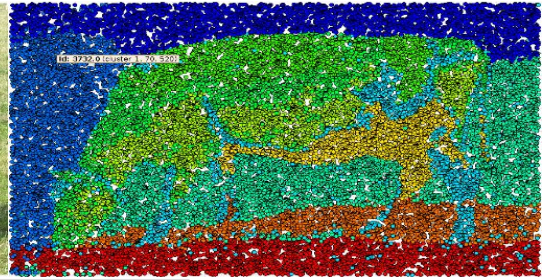
Aprendizaje Supervisado







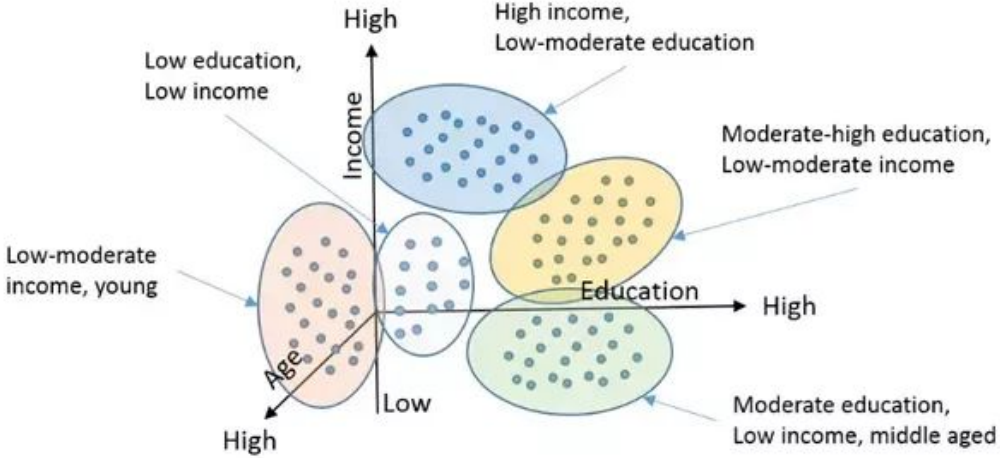


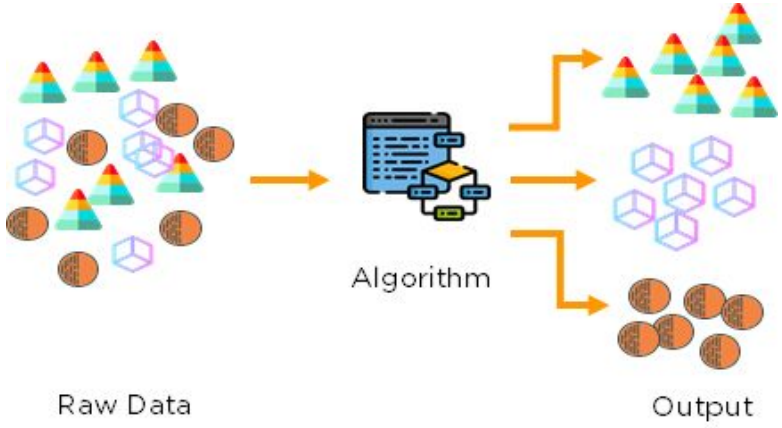


Aprendizaje Supervisado

Aprendizaje no Supervisado

Aprendizaje por Refuerzo



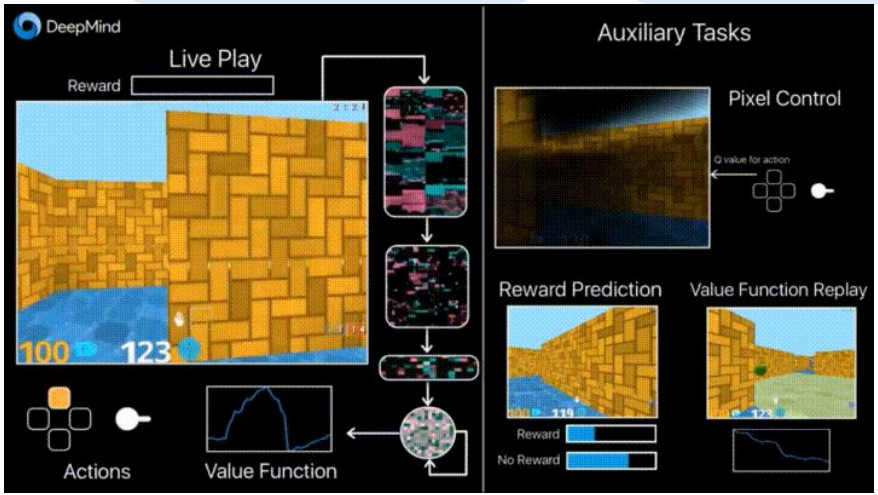


Machine Learning

Aprendizaje Supervisado

Aprendizaje no Supervisado

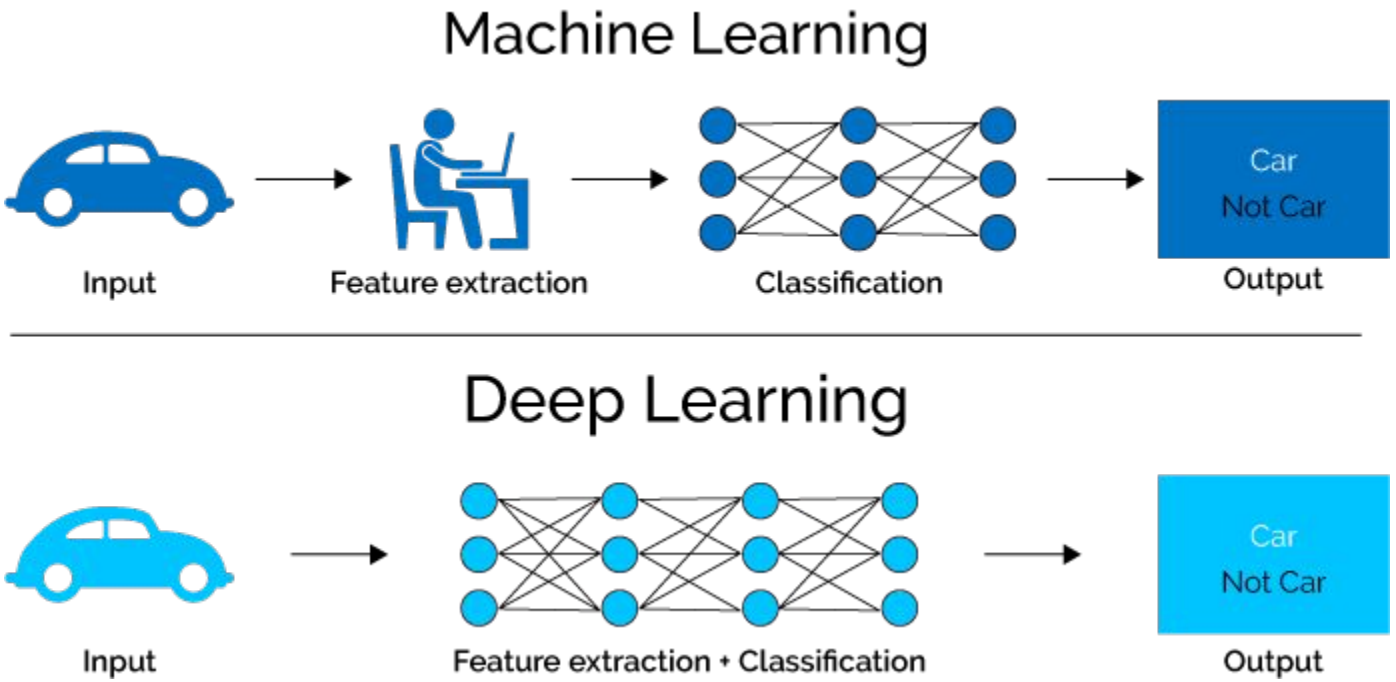
Aprendizaje por Refuerzo



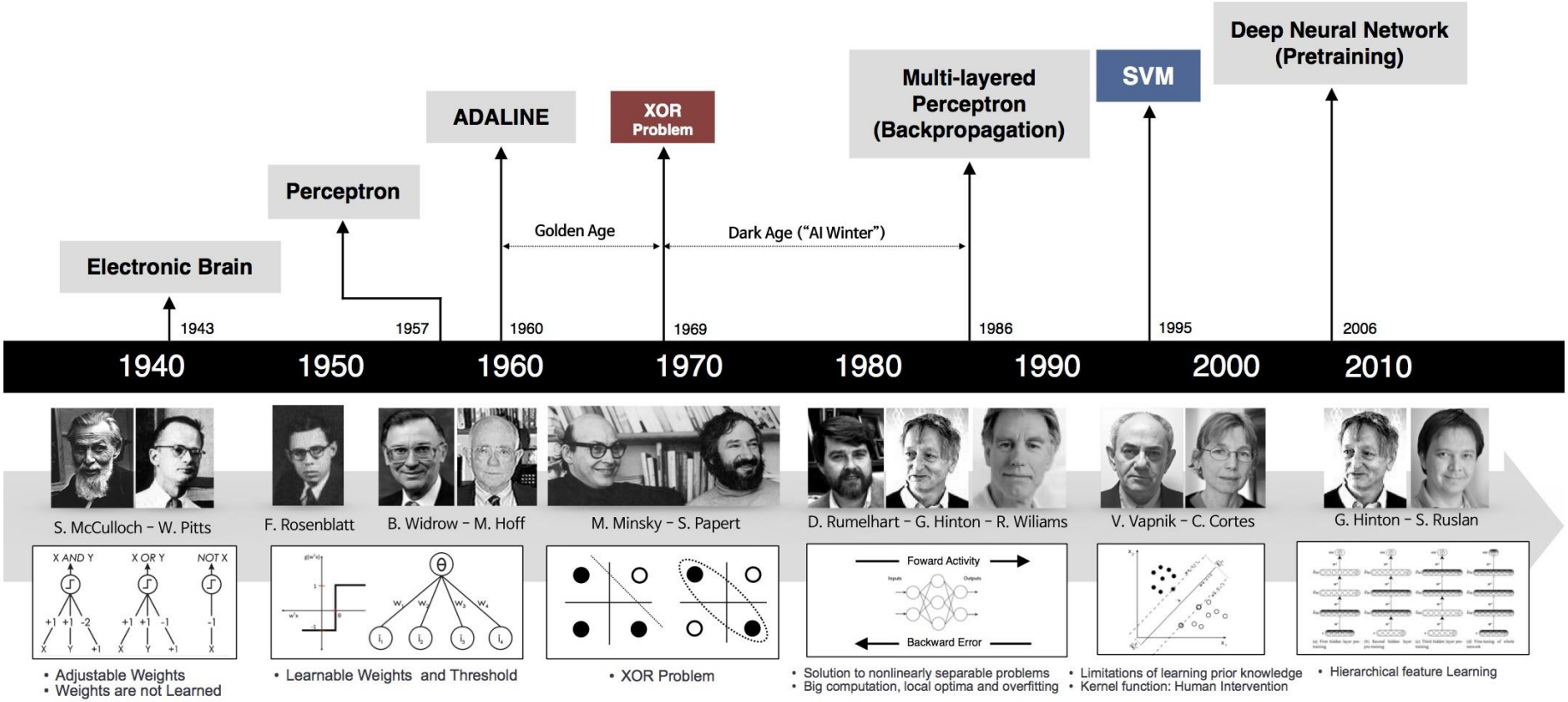


Deep Learning

From Machine Learning to Deep Learning



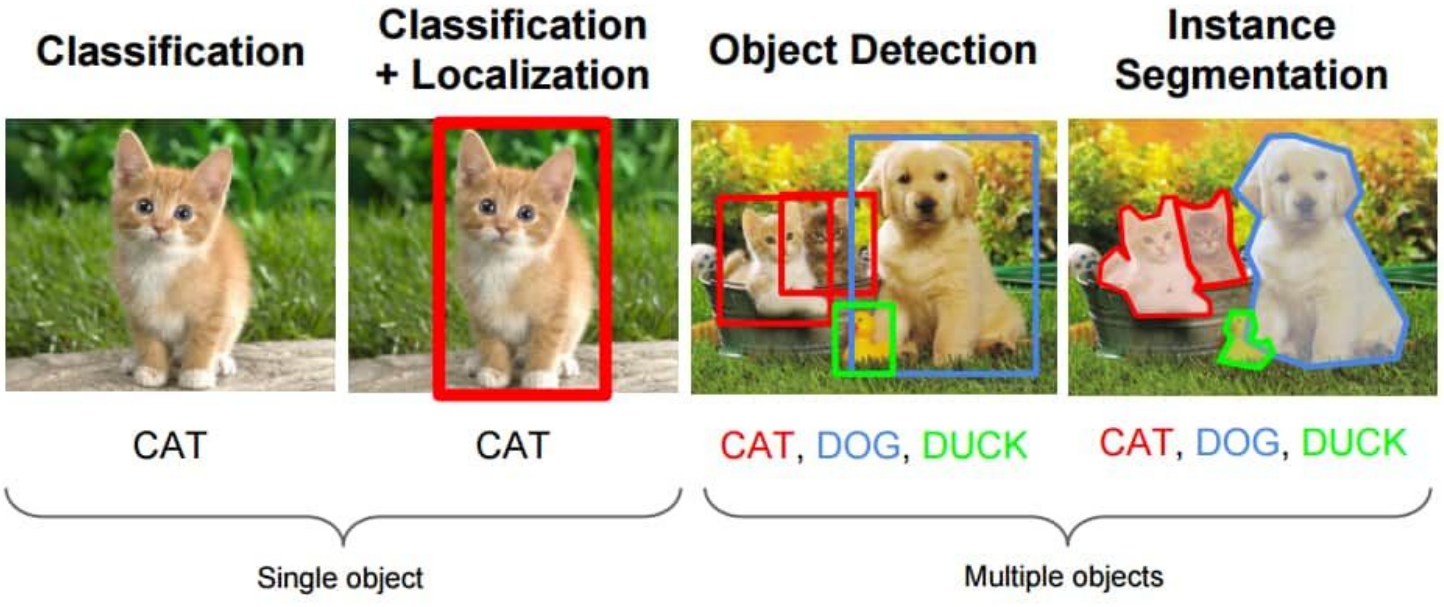

WOMAN
P


Deep Neural Network (Pretraining)
SVM
Multi-layered Perceptron (Backpropagation)
XOR Problem
ADALINE
Perceptron
Electronic Brain
Golden Age
Dark Age ("AI Winter")
1943
1957
1960
1969
1986
1995
2006
1940
1950
1960
1970
1980
1990
2000
2010
S. McCulloch - W. Pitts
F. Rosenblatt
B. Widrow - M. Hoff
M. Minsky - S. Papert
D. Rumelhart - G. Hinton - R. Wiliams
V. Vapnik - C. Cortes
G. Hinton - S. Ruslan
X AND Y
X OR Y
NOT X
Foward Activity
Backward Error
• Adjustable Weights
• Weights are not Learned
• Learnable Weights and Threshold
• XOR Problem
• Solution to nonlinearly separable problems
• Big computation, local optima and overfitting
• Limitations of learning prior knowledge
• Kernel function: Human Intervention
• Hierarchical feature Learning

¿Qué problemas se están resolviendo con DL?

- Visión computacional
  - Reconocimiento de imágenes
  - Segmentación de objetos
  - Reconstrucción de imágenes

Source: Fei-Fei Li, Andrej Karpathy & Justin Johnson (2016) cs231n, Lecture 8 - Slide 8, Spatial Localization and Detection (01/02/2016)



¿Qué problemas se están resolviendo con DL?

- Lenguaje natural (NLP)
  - Traducción
  - Parsing
  - Generación de conversaciones (chatbots)
  - Generación de resúmenes

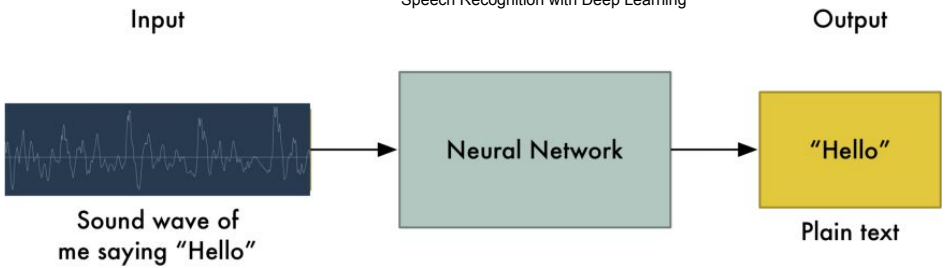


Sorce: Data Science Lab. Available: http://datascience.jd.com/?page_id=251

¿Qué problemas se están resolviendo con DL?

- Procesamiento de señales o datos en secuencia
  - Reconocimiento de acciones (sensores)
  - Reconocimiento de voz

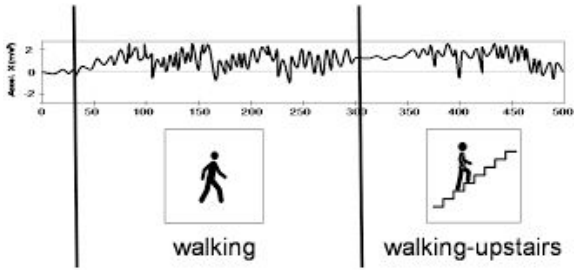
Source: Adam Geitgey. Machine Learning is Fun Part 6: How to do Speech Recognition with Deep Learning



- Tareas combinadas
  - Generación de descripciones de imágenes (Image captioning)
  - Reconocimiento de videos



Source. Ran Xu, Priyanshu Agarwal, Suren Kumar, Venkat N. Krovi, and Jason J. Corso Combining Skeletal Pose with Local Motion for Human Activity Recognition



Source: Rubén San-Segundo, Juan M. Montero, José Moreno-Pimentel, José M. Pardo. HMM Adaptation for Improving a Human Activity Recognition System

Image Captioning

Describes without errors	Describes with minor errors	Somewhat related to the image	Unrelated to the image
 A person riding a motorcycle on a dirt road.	 Two dogs play in the grass.	 A skateboarder does a trick on a ramp.	 A dog is jumping to catch a frisbee.
 A group of young people playing a game of frisbee.	 Two hockey players are fighting over the puck.	 A little girl in a pink hat is blowing bubbles.	 A refrigerator filled with lots of food and drinks.
 A herd of elephants walking across a dry grass field.	 A close up of a cat laying on a couch.	 A red motorcycle parked on the side of the road.	 A yellow school bus parked in a parking lot.

[Vinyals et al., 2015]

A flower with thin pink petals resting around a cluster of white and yellow stamen.



This flower is white and yellow in colour, with petals that are multi coloured.



A fat round bird with a bright yellow breast and blue on top of the head.



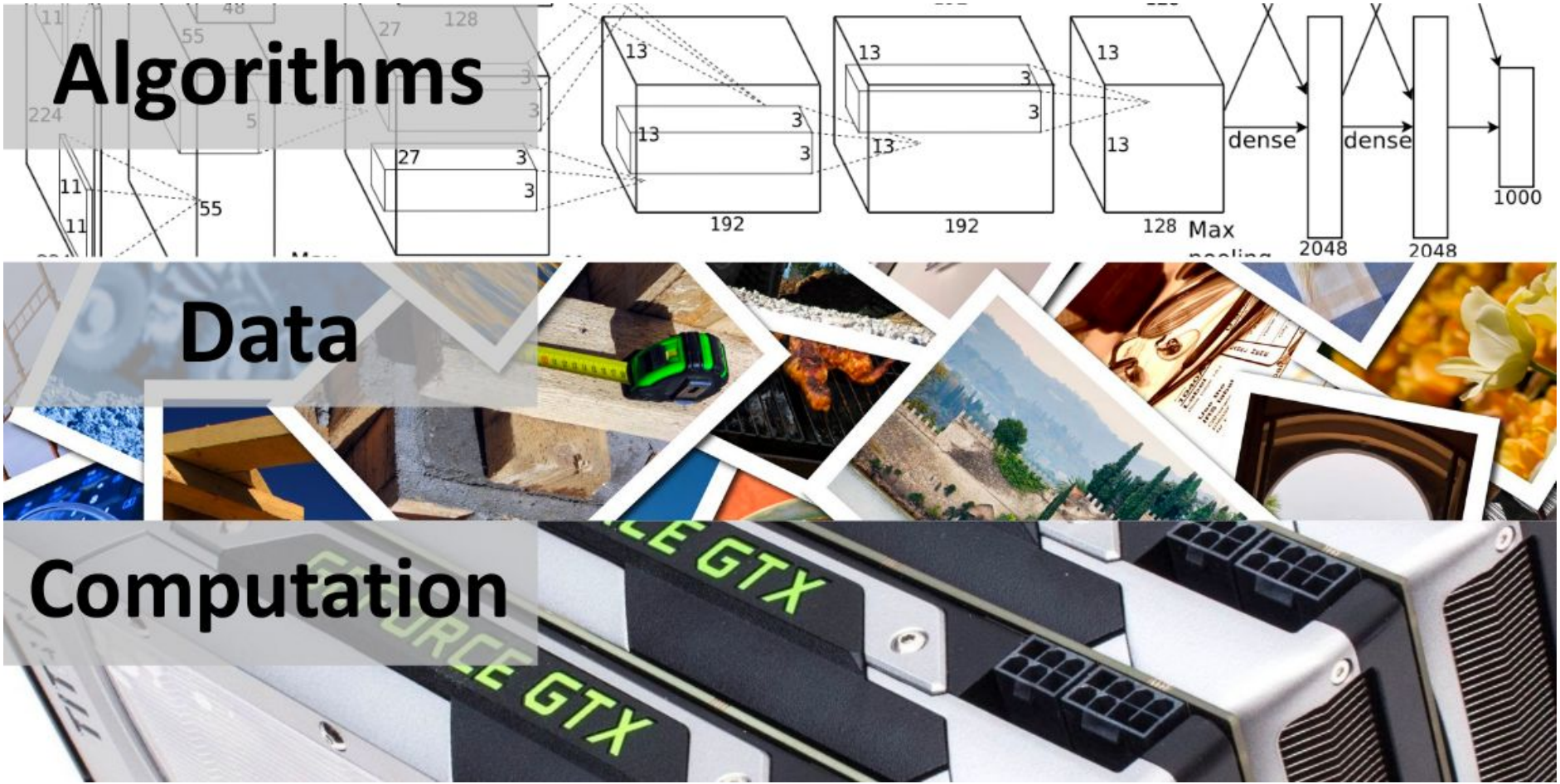
The tan bird has a short beak, with stone black eyes.



Text to Image Synthesis

[Reed et al., 2016]

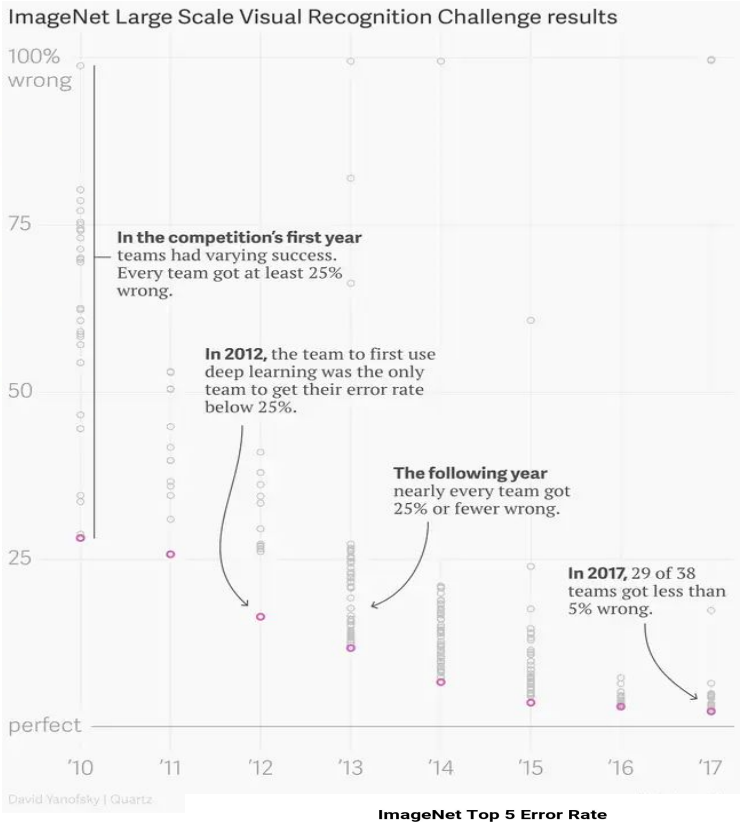
Deep Learning Ingredientes

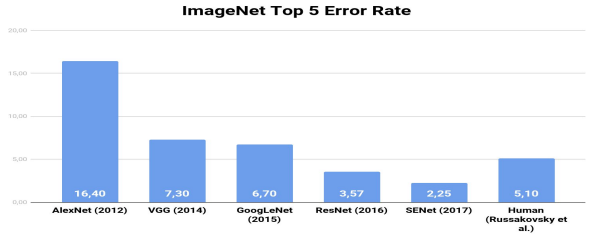

Algorithms
Data
Computation
dense
dense
1000
2048
2048
192
192
128
Max

IMAGENET

- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.







https://qz.com/1034972/the-data-that-changed-the-direction-of-ai-research-and-possibly-the-world

IA Generativa

Modelos generativos de texto a imagen

Midjourney

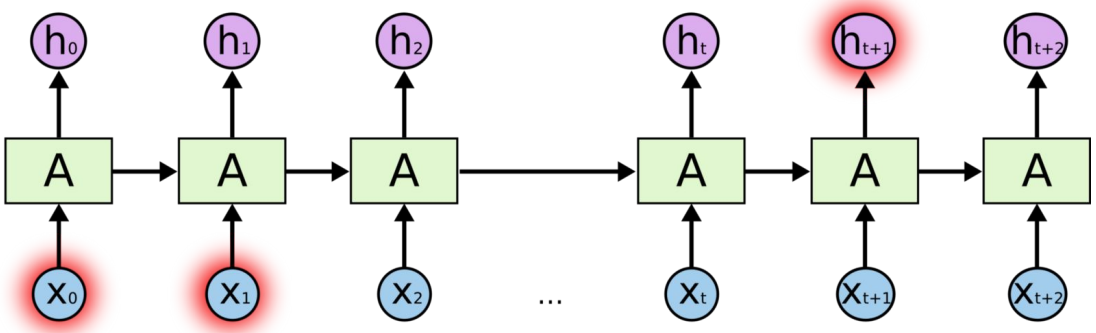
DALL·E 2

stability ai

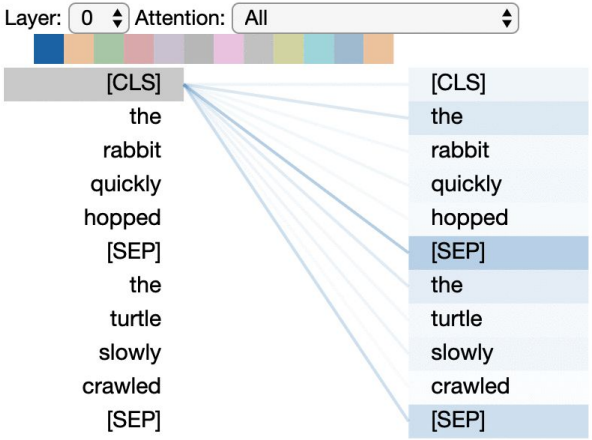


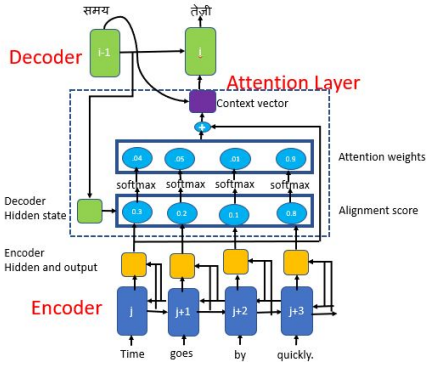
Problema de las RNN

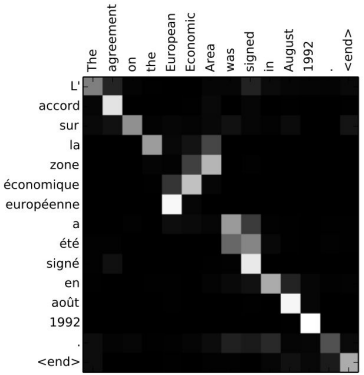
'Vivo en Brasil durante los últimos 20 años y trabajo como consultor durante los últimos 10 años 'Sé ____?____'.

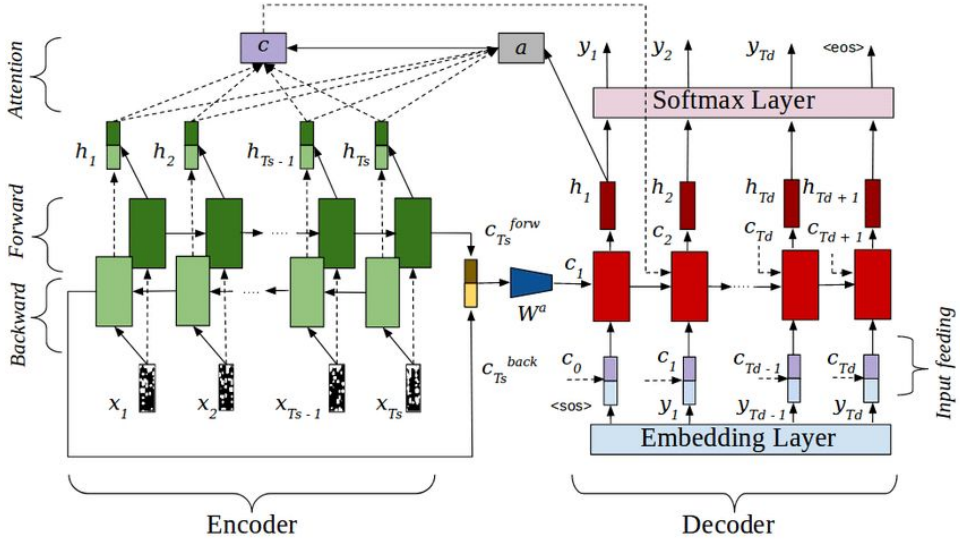


Redes de Atención!!!



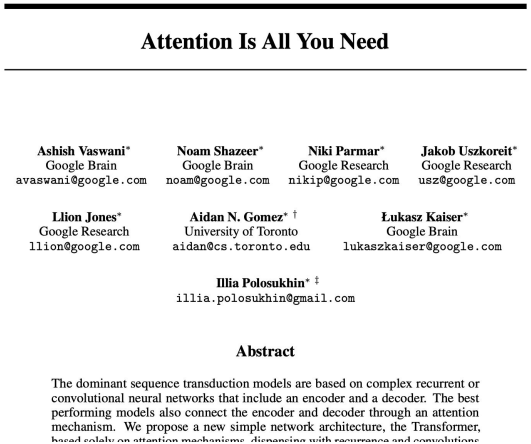






https://medium.com/subex-ai-labs/breakdown-the-bert-in-pieces-df46f60b65d8

Transformers!!



Attention Is All You Need

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Niki Parmar* Google Research nikip@google.com

Jakob Uszkoreit* Google Research usz@google.com

Llion Jones* Google Research llion@google.com

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

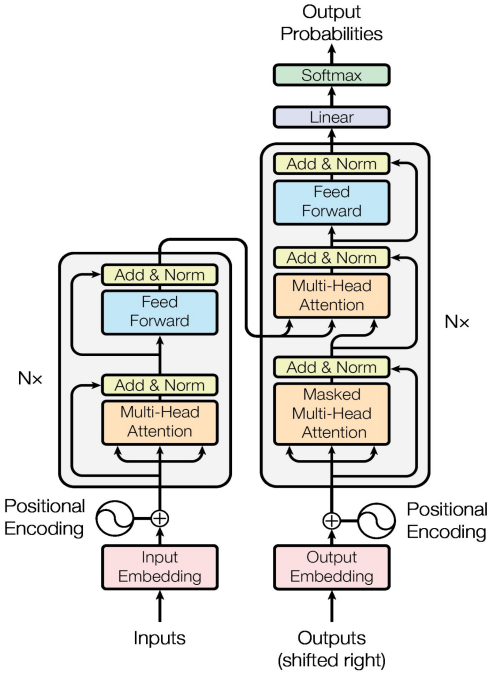
Illia Polosukhin* ‡ illia.polosukhin@gmail.com

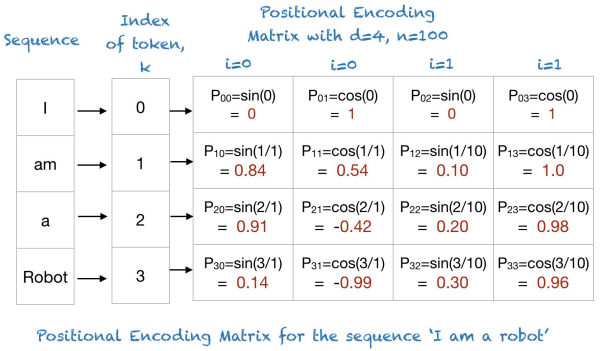
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions

v5 [cs.CL] 6 Dec 2017

https://arxiv.org/abs/1706.03762





Large Language Models (LLaMa)

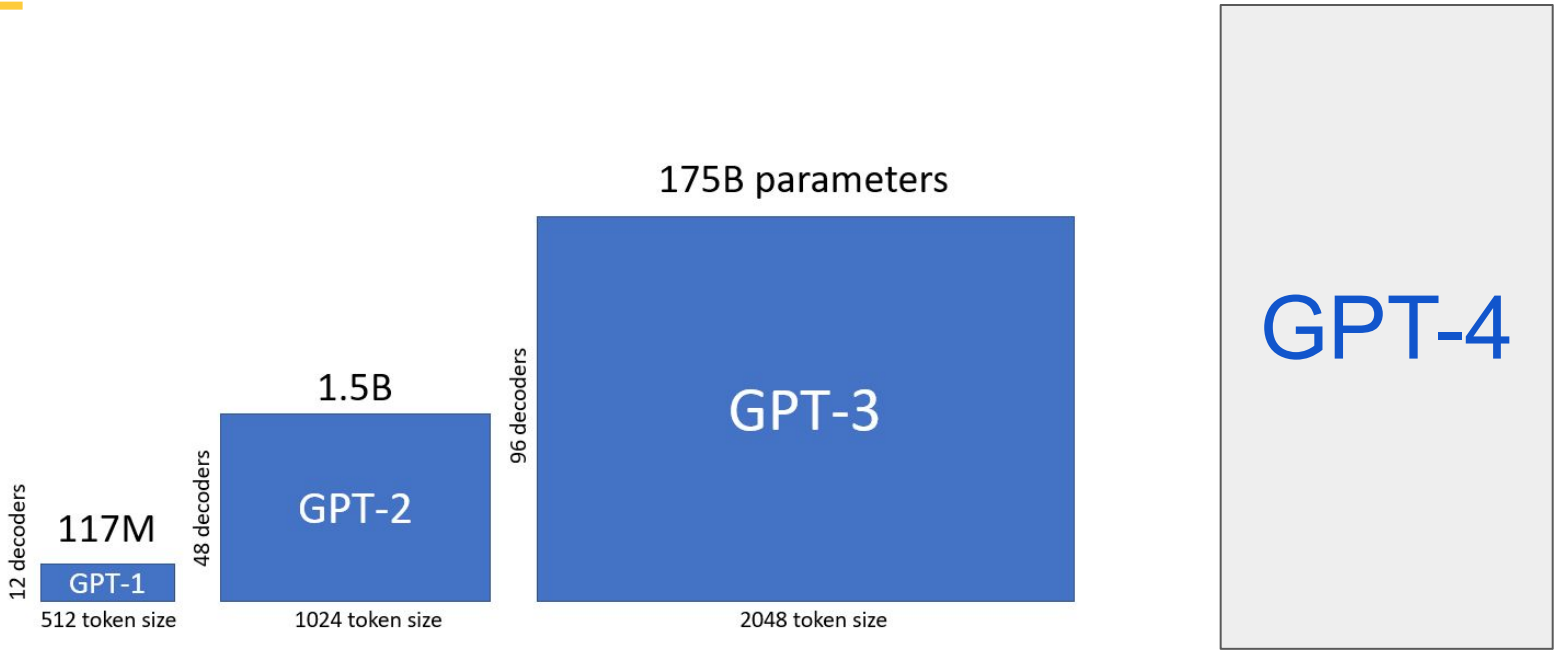


ChatGPT

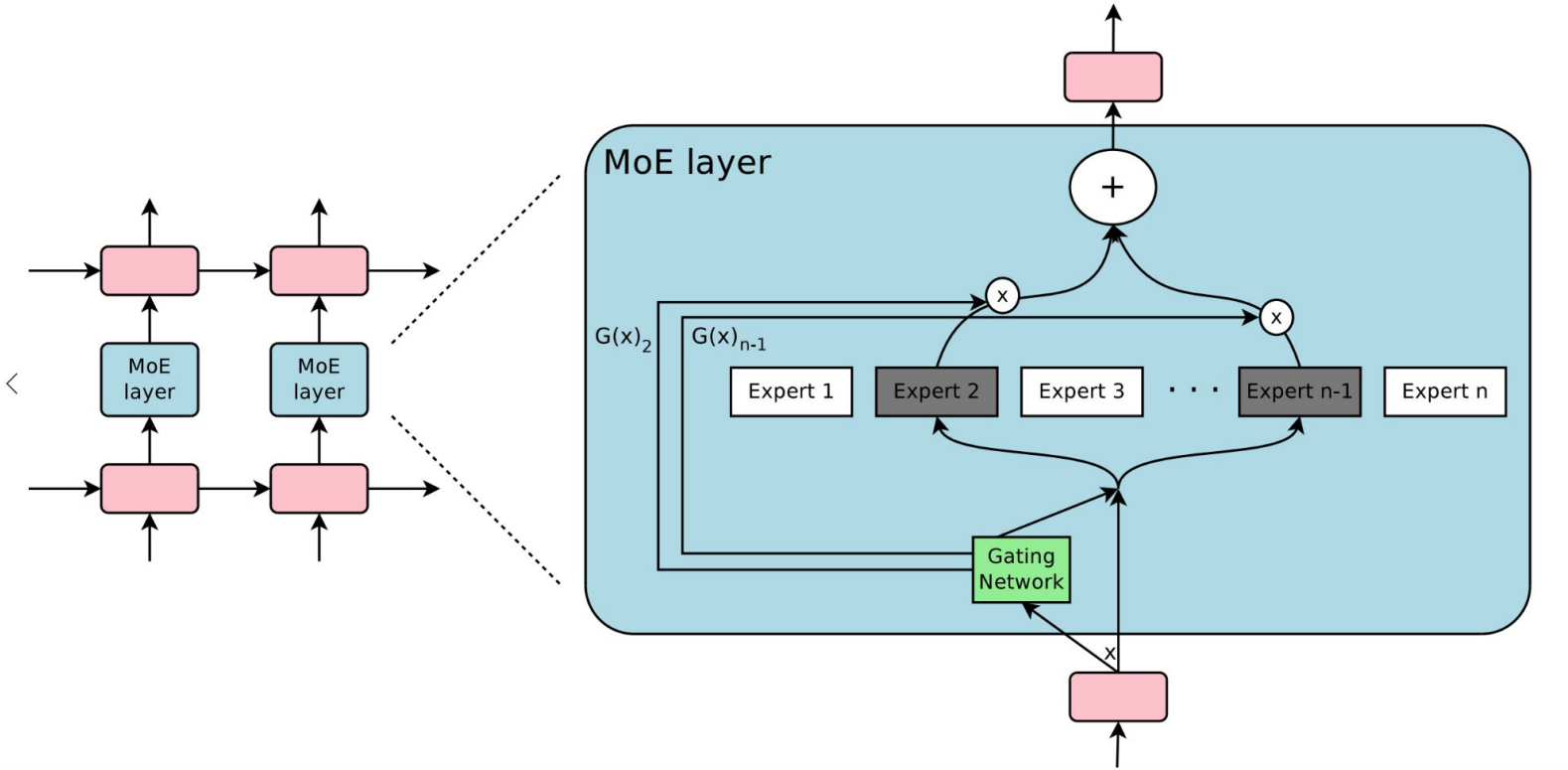
Bard

Stanford Alpaca

Generative Pre-trained Transformer (GPT)



GPT-4 Mixture of Experts





GRACIAS POR SU ATENCIÓN